VINU BALAN J

Software Engineer — Full-Stack & Applied AI | React, Next.js, TypeScript, Python, Node.js | LLM & RAG | Azure

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CAREER SUMMARY

Software Engineer with 3+ years at AT&T building production web applications, backend services and AI-powered features end to end. Builds responsive, high-performance interfaces with React, Next.js and TypeScript, and RESTful microservices with Python (FastAPI, Flask) and Node.js on Microsoft Azure with Docker and CI/CD. Develops and productionizes LLM applications, including Retrieval-Augmented Generation (RAG) with data ingestion, embeddings, vector search, query rewriting and real-time streaming, and trained a custom machine-learning translation service used in production. Experienced in A/B testing and experimentation (Adobe Target, Adobe Analytics), unit and end-to-end testing, performance tuning, code reviews and mentoring. Daily user of AI-assisted development tools (Claude Code, GitHub Copilot, Cursor), validating output through review, tests and secure coding standards. Winner, AT&T Bounty Sprintathon 2025 ($50K).

EDUCATION

Master of Business Administration — Artificial Intelligence & Data Science 2025 – Present

SRM Institute of Science and Technology, Kattankulathur, Chennai

Bachelor of Technology — Artificial Intelligence & Data Science 2020 – 2024

Chennai Institute of Technology, Chennai

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3, Java (working knowledge)

Frontend: React (Hooks, Context API), Next.js, Redux, Zustand, Tailwind CSS, SASS/SCSS, Styled Components, Responsive UI, Webpack, Vite, Chrome Extensions, Performance Optimisation (Lighthouse, Chrome DevTools)

Backend & APIs: Node.js, Express.js, FastAPI, Flask, Django, RESTful APIs, Microservices, JSON, API Design, Object-Oriented Design, Streaming Responses

AI, LLM & Generative AI: LLM APIs, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, Embeddings, Vector Search, ChromaDB, Ollama, Query Rewriting, Document Scoring, Real-time Streaming UIs

Machine Learning & Data: Machine Learning, NLP, Model Training & Evaluation, Feature Engineering, Pandas, NumPy, Scikit-learn, spaCy, OpenCV, Data Cleaning, Exploratory Analysis

Databases: PostgreSQL, SQL, MongoDB, Redis, ChromaDB

Cloud & DevOps: Microsoft Azure, Azure DevOps, AWS (fundamentals), Docker, Kubernetes (fundamentals), Jenkins, CI/CD, Git, Azure Monitor, Application Insights

Testing & Quality: Jest, React Testing Library, Playwright, Cypress, PyTest, Selenium, Unit, Integration & End-to-End Testing, Code Reviews, Secure Coding

Experimentation & Analytics: A/B Testing, Adobe Target, Adobe Analytics, Server-side Experimentation

AI-Assisted Development: Claude Code, GitHub Copilot, Cursor, Agentic Engineering Practices

Practices: Agile / Scrum, SDLC, Jira, Confluence, Mentoring, Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

Senior Associate Applications Developer — AT&T May 2024 – Present | Chennai, India

- Developed high-performance, responsive customer-facing applications with React, Next.js, TypeScript and JavaScript, integrated with backend REST APIs, contributing to scalable and maintainable production systems.
- Built ML-powered Python services on Microsoft Azure that replaced paid third-party API calls, lowering operational costs while improving application performance.
- Developed a scalable machine-learning translation service (English to Spanish): prepared training data, trained and evaluated custom models, and exposed them to production applications through REST APIs.
- Designed and implemented A/B testing and server-side experimentation with Next.js and Adobe Target, and analysed results in Adobe Analytics to drive data-informed product and campaign decisions.
- Built internal Chrome extensions, automation tools and Next.js web apps that removed third-party dependencies, generating cost savings and new revenue opportunities; workflow automation saved the team 80+ hours.
- Used AI-assisted development tools (GitHub Copilot, Claude Code, Cursor) for code generation, refactoring, unit tests and documentation, validating output through peer review and automated testing.

- Championed engineering best practices through code reviews, reusable components and maintainable architecture; mentored junior engineers.
- Shipped through CI/CD with Jenkins and Git in Scrum teams, working with product managers, designers and engineers to turn business requirements into production features.

Technologies: React, Next.js, TypeScript, JavaScript, Python, Node.js, Machine Learning, Microsoft Azure, REST APIs, Adobe Target, Adobe Analytics, Jenkins, Git

Staff Associate Technical Intern — AT&T May 2023 – May 2024 | Chennai, India

- Developed modular, reusable React components supporting multiple product surfaces in production applications.
- Translated product requirements into production-ready features with product managers and designers.
- Analysed user and product data in Adobe Analytics to identify improvements and optimisation opportunities.
- Strengthened CI/CD workflows with Jenkins and Git; tracked delivery in Jira.

Technologies: React, Next.js, JavaScript, Adobe Analytics, Git, Jira, Jenkins

Research Intern — Buddi.ai June 2021 – December 2021

- Developed an AI-powered resume-parsing platform in Python (Flask, spaCy, NLP, Machine Learning) to extract structured information from unstructured resumes.
- Designed a graph-based filtering algorithm that processed millions of resumes for scalable candidate filtering and matching.

Technologies: Python, Flask, NLP, spaCy, Machine Learning

PROJECTS

AI-Powered Government Schemes Assistant — Full-Stack RAG Application

Technologies: Next.js, TypeScript, Tailwind CSS, Python, FastAPI, LangChain, ChromaDB, Ollama, PostgreSQL, Redis, Docker, Jest, Playwright

- Built an end-to-end Retrieval-Augmented Generation (RAG) application: a FastAPI microservice with streaming endpoints and a Next.js / TypeScript conversational UI that renders real-time streaming responses.
- Built a data ingestion pipeline to crawl, extract, clean and chunk content from Tamil Nadu and Government of India websites, indexing embeddings in ChromaDB.
- Improved answer relevance with query expansion, document scoring and retrieval filtering over a locally hosted LLM (Ollama, LangChain).
- Stored metadata in PostgreSQL, cached frequent queries in Redis, containerized with Docker, and added structured logging and health checks; managed chat state with React Context and hooks; tested with Jest, React Testing Library and Playwright.

Product Experimentation & A/B Testing Platform

Technologies: React, Next.js, JavaScript, Node.js, REST APIs, Adobe Target

- Built an internal platform with a Node.js REST backend and React / Next.js dashboards so product teams could configure, run and analyse A/B experiments without engineering support.
- Developed dashboards for experiment configuration and results visualization to support data-driven product decisions.

Advanced AI Resume Parser & Search Engine

Technologies: Python, NLP, Django, MongoDB, JavaScript, Bootstrap 5

- Built an NLP-powered resume parsing and search web application with MongoDB data models and an interactive JavaScript interface for candidate discovery.

PC Automation Using Hand Gestures

Technologies: Python, OpenCV, Machine Learning, Flask, JavaScript

- Built real-time hand-gesture detection and classification with OpenCV and machine learning, with a Flask backend and a browser-based monitoring interface.

ACCOMPLISHMENTS

- AT&T Bounty Sprintathon 2025 — Winner ($50K cash prize)
- AT&T Transformer Award | AT&T G.O.A.T Award | AT&T Service Excellence Award
- Smart India Hackathon — Winner (₹1 Lakh cash prize); Cognizant Prodigy Hackathon — Winner; Tech Zeal Hackathon — Winner